

The Food Sector: Environment and Sustainability

Science

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The Food Sector: Environment and Sustainability (A33361)

Short Title: Food Environ & Sustainability
Faculty Science and Computing
Department Science
Cluster Environmental Science
Author EDUGGAN
Credits 5

Level: Advanced

Description of Module / Aims

This module examines the impact the food industry has on our environment. The waste management hierarchy is examined in detail as well as an introduction to the challenges facing the food industry. The value of food wastes is examined. The waste management strategy and environmental impact of a food company are investigated as part of a case study. The conversion of food waste into energy is studied through an introduction to a variety of newly established technologies.

Programmes

F00D-0031 BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)

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Indicative Content

- Effects of food industry pollution on the environment
- Food security and food waste and the circular bioeconomy
- Types of food waste
- Waste management hierarchy model
- Food waste as a resource, food waste to energy technologies
- Greenhouse gases and carbon footprint
- Legislation relating to food waste / processing and the environment
- Energy conservation and recovery within the food sector

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Assess of the causes and effects of pollution from the food industry.
2. Assess the future impact of food security and sustainability.
3. Establish the different food waste types, pre- and post-consumer.
4. Critique the waste management hierarchy model within the circular bioeconomy.
5. Evaluate the importance of greenhouse gas emissions and the carbon footprint from farm to retail food product.
6. Appraise the legislation associated with the management of food wastes.
7. Assess energy conservation and recovery using waste to energy conversion technologies within the food sector.

Learning and Teaching Methods

- Lectures on the stated indicative content will be delivered.
- Field trips will reinforce and support theoretical elements of the course.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	2,3,4,5,6,7
Continuous Assessment	40%	
Case Studies	40%	1,2,3,6,7

Assessment Criteria

<40%: No understanding of basic principles of environmental aspects of food production and food security.

40%-49%: Able to understand principles; may have some difficulty applying this theory.

50%-59%: Able to understand principles discussed in the course and also able to evaluate practical data and evaluate food processing from an environmental point of view.

60%-69%: In addition to the above, able to demonstrate a high level of competence and efficiency in subject area.

70%-100%: All previous to excellent level. Be able to describe and discuss in detail environmental aspects of food production, processing and food security.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	30	
Field Trips	6	
Independent Learning	99	

Essential Material

- Arvanitoyannis, I.S. *_Waste Management for the Food Industries_*. London: Elsevier Academic Press, 2008.
- Yadav, S.S., R.J. Redden, J.L. Hatfield, A.W. Ebert and D. Hunter. *_Food Security and Climate Change_*. Chichester, West Sussex, England: Wiley Blackwell, 2019.

Supplementary Material

- "Department of the Environment, Community and Local Government." <http://www.environ.ie/en/>
- "The European Commission - A European Green Deal." https://ec.europa.eu/info/strategy/priorities-2019-2024/european-green-deal_en
- "The United Nations Climate Action - What is Climate change." <https://www.un.org/en/climatechange/what-is-climate-change>
- Tiwari, B.K., T. Norton and N.M. Holden. *_Sustainable Food Processing_*. England: Wiley-Blackwell, 2013.